WASP 2026 · IJCNLP-AACL 2026 · ASTROCLIMB SHARED TASK

Beyond Visual Similarity

Parameter-Efficient Vision–Language Models for Scientific Citation Linking

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≈0.74
test Macro-F1 · fine-tuned Qwen3.8-27B
(4-bit LoRA, 0.45% of parameters)

01 The task

The AstroCLIMB shared task evaluates whether multimodal models can identify citation relationships among astronomy papers from pairs of figures and captions.

```
ŷ = argmax P(y = c | x₁, x₂, τ)
τ ∈ {ii, ti, tt}

ii: Image–Image modality
ti: Text–Image modality
tt: Text–Text modality
```

Same figure	Composite panels, true figure–caption pairs
Same paper	Distinct artifacts from one paper
Related papers	Linked by citations or topical overlap
Unrelated papers	Independent publications

02 Dataset

Split	ii	ti	tt	Total
Train	2,550	3,400	2,550	8,500
Val	450	600	450	1,500
Test	3,000	4,000	3,000	10,000

Labeled pairs: 1,000 same figure and 3,000 for each other tier. Stratified train/val split keeps class and modality mix; test labels are blind (Kaggle).

03 Model progression

- **E0** **Frozen baselines.** Cosine similarity of OpenAI / SigLIP embeddings, thresholds fitted on train.
- **E1** **Bi- and cross-encoders.** AstroBERT + ViT / SigLIP-1/2 (top 4 blocks unfrozen); PaliGemma.
- **E2** **Generative VLMs.** Qwen3-VL-8B / 32B and Qwen3.8-27B, tuned with Unsloth LoRA.
- **E3** **Specialist and cascade.** 2-class same-paper vs related adapter; two-level hierarchy.

04 Best system: pipeline

INPUT Pair (x_1, x_2) + modality τ → **MODEL** Qwen3.8-27B, 4-bit NF4 + LoRA → **DECODE** First-token logits, 4-way softmax → **CALIBRATE** A↔B averaging + logit offsets

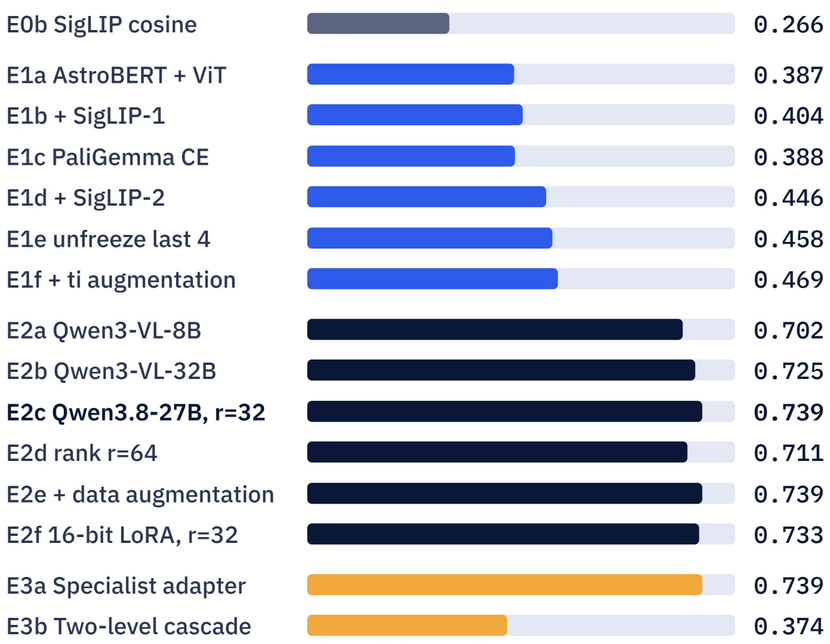
0.45%
124.4M of 27.4B parameters trained

9.5 h
GPU time, 10 epochs (RTX PRO 6000)

~1 h
to infer 10,000 test pairs

- LoRA on Wq, Wk, Wv, Wo; rank r ∈ {32, 64}, α = 16, AdamW 8-bit, lr 1×10^{-4}, batch 4 × 8 accumulation.
- Text–Image upsampling (weakest modality in initial cross-encoder validation, Macro-F1 0.3604) and A↔B pair-order permutation for symmetric invariance.
- Next-token logit classification eliminates autoregressive generation latency.

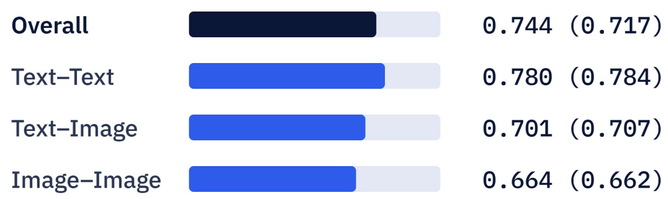
05 Test Macro-F1 by experiment



Qwen3.8-27B outperforms both Qwen3-VL-8B and the nominally larger Qwen3-VL-32B. LoRA rank, 16-bit precision and augmentation all perform within a narrow score band.

06 Text beats figures

Validation Macro-F1 (accuracy) for Qwen3.8-27B, by input modality:



- **tt:** benefits from explicit domain terminology (e.g., coordinates and instruments).
- **ti:** effectively grounds text into visual elements.
- **ii:** shared plot conventions (contours, spectra, colour–magnitude diagrams) create spurious similarity.

06b Where it breaks

Same figure and unrelated papers are classified reliably. Most errors sit on one boundary: **same paper vs related papers** — papers that cite each other or study the same object share vocabulary and plot styles.

Specialist adapter vs generalist: virtually identical, within run-to-run variance.

0.374
Uncalibrated cascade: stage-1 mF1 of 0.55 discards pairs early.

07 Limitations

Lack of document context. Without surrounding text, methodology or author metadata, same paper vs related papers is very difficult.

Visual domain gap (Image–Image). Standard vision models are not trained on astronomy plots; encoders focus on shared plot styles.

Simple decision rule. A simple argmax rule; hard multi-stage decisions fail easily, as the cascade (E3b) showed.

Inference speed and scalability. A 27B model is slow and resource-heavy, hard to scale across millions of NASA ADS papers.

Lack of explanations. First-token prediction is fast but cannot explain why two items are linked.

Domain focus. Findings are specific to astronomy and may not apply to biomedicine or computer science.

Future work: full-paper metadata, domain-specific vision encoder pretraining, and dedicated decision models (e.g., KEV/JEV) for routing.

08 Takeaways

Parameter-efficient VLMs work. LoRA on 0.45% of weights reaches ≈0.74 test Macro-F1.

Looks are not links. Visual similarity alone is an insufficient proxy for citation linkage (tt > ti > ii).

Not capacity-limited. Performance is limited less by architecture than by Image–Image ambiguity and the same-paper vs related-papers boundary.

Keywords: multimodal learning · VLM · citation linking · PEFT

Benchmark: AstroCLIMB (Grezes et al., 2026), NASA ADS

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